

POSEY FIELD, AL, USA

WMO: 720413

Lat: 34.267N

Lon: 87.600W

Elev: 932

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	20.5	3.3	6.8	22.3	8.9	9.0	25.2	19.2	47.3	18.0	48.4	5.2	0	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	93.2	75.4	91.4	75.8	90.2	75.5	79.0	88.7	78.1	87.7	77.2	86.6	5.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	142.7	83.2	75.3	137.3	82.1	74.5	133.6	81.2	43.0	88.7	41.9	88.5	41.0	86.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.1	15.2	12.7	9.5	96.6	5.5	3.0	5.5	98.8	2.3	100.5	-0.8	102.2	-4.8	104.4		
			8.4	81.1	5.4	0.9	4.5	81.7	1.3	82.2	-1.7	82.7	-5.6	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	39.7	44.6	51.8	60.7	68.3	76.6	78.5	77.5	72.7	61.2	49.1	44.3
	DBStd	15.88	11.28	11.17	9.83	7.76	6.87	4.11	3.94	3.83	6.00	8.54	8.85	10.11
	HDD50	1061	355	220	101	11	1	0	0	0	0	12	120	241
	HDD65	3310	784	575	415	171	51	1	0	0	12	176	481	644
	CDD50	4886	36	68	157	332	567	797	885	851	680	358	92	63
	CDD65	1662	0	3	7	42	152	348	420	386	242	57	4	1
	CDH74	14926	0	5	65	388	1292	3338	3877	3270	2128	540	22	1
	CDH80	6119	0	0	7	51	384	1488	1768	1403	873	142	3	0
Wind	WSAvg	5.0	6.1	6.6	6.6	6.3	4.9	3.7	3.4	3.1	3.2	4.3	5.2	6.0
Precipitation	PrecAvg	60.9	5.8	5.9	5.6	5.2	5.2	5.2	5.2	4.0	3.9	3.4	5.1	6.1
	PrecMax	79.5	10.5	10.5	10.0	13.4	14.6	13.5	9.1	8.7	6.6	6.9	9.5	16.9
	PrecMin	36.6	2.2	2.6	0.8	1.5	0.7	1.4	1.8	0.2	0.6	0.0	3.1	2.5
	PrecStd	10.4	2.3	2.5	2.2	2.8	3.8	2.8	2.2	2.1	1.6	2.0	1.7	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	74.2	81.0	83.9	90.2	97.0	96.9	96.8	94.2	89.0	76.7	71.3
		MCWB	61.3	64.8	64.1	66.2	73.0	72.2	75.9	78.4	72.0	72.4	61.5	65.3
	2%	DB	64.2	70.0	75.0	81.4	87.0	92.5	93.0	92.5	91.2	83.6	72.0	67.9
		MCWB	60.4	62.6	61.9	65.3	71.9	75.6	76.6	76.2	73.0	69.1	59.4	64.2
	5%	DB	61.2	66.2	71.9	79.0	84.1	90.6	91.1	90.4	88.9	80.4	67.9	64.3
		MCWB	57.0	60.5	60.4	64.3	70.2	75.6	76.7	76.0	73.5	66.5	60.6	60.7
	10%	DB	57.2	62.6	68.0	75.3	81.7	88.4	89.7	88.0	85.5	76.0	64.3	61.2
		MCWB	53.0	55.7	58.2	62.8	69.5	74.7	76.3	75.4	72.9	65.5	57.1	57.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	67.9	67.1	70.7	75.1	79.3	80.3	80.4	77.7	75.1	68.5	67.6
		MCDB	65.8	71.1	74.6	78.8	84.6	89.1	89.6	90.3	85.6	82.5	70.2	69.4
	2%	WB	61.6	65.2	64.8	68.5	73.6	78.1	79.0	79.0	76.2	73.3	65.5	64.3
		MCDB	63.0	68.5	71.2	76.2	83.2	88.5	88.7	88.7	85.1	80.1	68.4	66.4
	5%	WB	58.6	62.2	62.9	67.1	72.3	77.2	78.2	77.9	75.1	70.5	62.2	61.9
		MCDB	60.8	65.2	68.5	74.6	81.5	87.6	87.8	87.0	84.0	76.1	64.6	64.0
	10%	WB	54.6	58.2	60.8	65.4	71.0	76.0	77.3	76.8	74.0	67.8	58.2	58.4
		MCDB	56.4	61.1	66.5	71.7	79.3	85.2	86.7	84.9	82.2	73.7	64.0	60.9
Mean Daily Temperature Range	5% DB	MDBR	19.2	19.7	21.9	23.2	21.1	21.7	19.9	19.5	22.1	23.5	22.9	17.8
		MCDBR	20.7	22.5	25.7	25.8	24.1	24.4	22.6	23.2	25.9	27.3	26.4	18.7
		MCWBR	16.1	14.6	14.8	11.7	10.4	9.6	8.3	9.0	10.1	12.9	16.3	14.7
	5% WB	MCDBR	16.7	17.5	20.8	19.9	20.2	21.1	20.0	19.2	20.7	20.2	18.5	17.1
	MCWBR	15.4	13.5	13.6	10.7	9.7	9.5	8.3	8.4	9.2	11.7	15.2	15.3	
Clear-Sky Solar Irradiance	taub	0.312	0.317	0.354	0.386	0.426	0.454	0.495	0.503	0.416	0.362	0.331	0.312	
	taud	2.520	2.503	2.422	2.320	2.252	2.196	2.110	2.081	2.315	2.451	2.480	2.536	
	Ebn at Noon	286	296	292	285	274	265	253	249	268	276	275	279	
	Edh at Noon	26	30	35	40	44	46	50	51	38	31	27	24	
All-Sky Solar Radiation	RadAvg	798	961	1328	1705	1878	1986	1903	1759	1536	1239	908	681	
	RadStd	64	125	122	120	142	157	110	104	135	161	95	68	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (31 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+169	

Nomenclature: See separate page